

Code-Layout, Readability and Reusability

Date	01 October 2026
Team ID	SWTID-2026-8336
Project Name	EduGenie — Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent indentation and formatting are maintained throughout the code.
2	Proper File Structure	Files and folders are logically organized	Yes	Frontend, backend, API, database, and supporting components are organized into separate files and folders.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables and functions use clear and meaningful names based on their purpose.
4	Function / Method Names	Functions are descriptively named	Yes	Functions and methods are named descriptively to clearly represent their operations.
5	Code Comments	Inline and block comments explain logic	Yes	Comments are used where necessary to explain important logic and implementation details.
6	Modular Design	Code is split into reusable functions/modules	Yes	Authentication, AI integration, learning features, database operations, and UI components are organized into reusable modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Repeated logic is minimized by using reusable components and functions.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Appropriate error handling is implemented for authentication, API requests, AI responses, and database operations.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	JavaScript, Node.js, Express.js, JWT, bcrypt	Login, Signup, Forgot Password and protected user access	High
2	AI Learning Assistant Module	JavaScript, Google Gemini API	Doubt clarification, simple explanations, question generation and learning assistance	High
3	Quiz Module	JavaScript, React.js, Node.js	Quiz generation, quiz attempts, results and answer evaluation	High
4	Progress Tracking Module	React.js, Node.js, MongoDB	Learning history, study plans, quiz results and progress tracking	High
5				

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	The project follows a structured frontend and backend organization with separate modules for major functionalities.
Readability	5	Clear naming, consistent formatting and modular implementation improve code readability.
Reusability	5	Reusable components and modules are used for authentication, AI learning, quizzes and progress tracking.
Documentation / Comments	4	Important implementation logic and major modules are documented with appropriate comments and descriptions.
Overall Score	5	The codebase is organized, readable and designed with reusable modules for future enhancement.